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ABSTRACT

Breast cancer is among the most prevalent malignancies affecting women worldwide and represents a significant healthcare burden. Effective treatment planning requires oncologists to analyze multiple **clinical parameters**, including **tumor characteristics**, **molecular biomarkers**, **staging information**, **lymph node involvement**, and **patient comorbidities**. However, these critical data points are often distributed across multiple diagnostic documents such as **histopathology reports**, **immunohistochemistry (IHC) panels**, **radiology findings**, and **laboratory tests**. This **fragmentation of clinical information** increases the complexity of decision-making and places a substantial **cognitive burden on healthcare professionals**.

As oncology treatment pathways become increasingly complex and personalized, there is a growing need for **intelligent systems** that can consolidate patient data and support clinicians in aligning treatment decisions with **established medical guidelines**. **Clinical Decision Support Systems (CDSS)** have been developed to address this need by analyzing patient information and generating **evidence-based recommendations**. However, many existing CDSS platforms face key limitations, including **limited transparency**, **high implementation costs**, and reliance on **opaque machine learning models** that do not provide clear reasoning for their outputs. The lack of **explainability** in such systems reduces **clinician trust** and limits their adoption in real-world clinical settings.

To address these challenges, this project proposes an **AI-Assisted Clinical Decision Support System for Breast Cancer Treatment Planning** that emphasizes **explainable and guideline-driven decision making**. The system integrates **Natural Language Processing (NLP)** techniques with a transparent **rule-based clinical reasoning engine** derived from internationally recognized oncology guidelines, including **NCCN, ESMO, ASCO, and ISMPO**. It automatically extracts **clinically relevant information** from diagnostic reports, transforms it into **structured patient profiles**, and evaluates the data using explicit **IF-THEN clinical rules**. Each recommendation is accompanied by a clear explanation linking the decision to specific **clinical parameters** and **guideline references**, ensuring full **traceability** and **interpretability**.

To further enhance safety and reliability, the system incorporates a **clinical data validation module** that flags **missing or uncertain information** for clinician review before generating recommendations. In addition, a **contraindication detection module** evaluates potential treatment risks based on **patient comorbidities** and **therapy-specific toxicity considerations**. The final treatment pathways are presented through a doctor dashboard, where clinicians can review the structured data, inspect the reasoning behind each recommendation, and modify or override system outputs, thereby maintaining full clinical control.

Given the sensitive nature of healthcare data, the system is designed with strong security and privacy safeguards, including encrypted data storage and transmission, role-based access control (RBAC), audit logging, and privacy-by-design principles aligned with HIPAA standards. By combining automated data extraction, explainable clinical reasoning, and secure system architecture, the proposed solution aims to improve the efficiency, transparency, and reliability of breast cancer treatment planning while reinforcing clinician trust and preserving human oversight in the decision-making process.

INTRODUCTION

Breast cancer represents one of the most significant public health challenges globally and is the **most common cancer among women in India**, accounting for approximately **27–30% of all female cancers**. Recent estimates indicate that **1 in 28 women in India is likely to develop breast cancer during her lifetime**, with higher incidence observed in urban populations. This rising burden underscores the need for efficient, accurate, and scalable approaches to treatment planning within the Indian healthcare context.

Advances in oncology research have established that breast cancer is not a single disease but a **heterogeneous group of malignancies** characterized by diverse molecular and biological profiles. These variations significantly influence **tumor behavior, treatment response, and patient prognosis**, making **precision medicine** central to modern breast cancer management. Treatment strategies are therefore increasingly tailored based on the unique clinical and molecular characteristics of each patient.

A critical component of this personalized approach is the analysis of **molecular biomarkers**, typically obtained through **immunohistochemistry (IHC)** testing. The most clinically relevant biomarkers include **Estrogen Receptor (ER)**, **Progesterone Receptor (PR)**, **Human Epidermal Growth Factor Receptor 2 (HER2)**, and the **Ki-67 proliferation index**. ER and PR determine **hormone sensitivity**, HER2 indicates **tumor aggressiveness and eligibility for targeted therapy**, and Ki-67 reflects **tumor proliferation rate**. While additional biomarkers such as **p53 mutations, BRCA1/2 genetic markers, and genomic assay scores (e.g., Oncotype DX)** are used in advanced settings, this project focuses on **ER, PR, HER2, and Ki-67** due to their **widespread availability, standardization, and routine use in Indian clinical practice**. This ensures that the proposed system remains **practical and applicable in real-world healthcare settings**, particularly in resource-constrained environments.

These biomarkers are used to classify tumors into **molecular subtypes** such as **Luminal A, Luminal B, HER2-enriched, and Triple-Negative Breast Cancer (TNBC)**, each associated with distinct biological characteristics and treatment pathways. In parallel, oncologists evaluate the **Tumor-Node-Metastasis (TNM) staging system**, which incorporates **tumor size, lymph node involvement, and distant metastasis**. Among these, **lymph node status is a critical prognostic factor**, as it indicates whether cancer has spread beyond the primary tumor. The number and extent of affected lymph nodes directly influence **disease stage, recurrence risk, and treatment decisions**, including the need for **chemotherapy, radiation therapy, and surgical intervention** such as axillary lymph node dissection.

Despite advances in diagnostic technologies, a major challenge in oncology practice remains the **fragmentation of clinical data**. Patient information relevant to treatment planning is often distributed across multiple reports generated by different diagnostic departments. **Pathology reports, imaging studies, biomarker test results, and laboratory findings** are frequently stored in separate systems, requiring clinicians to manually review and synthesize information before making treatment decisions. This process is time-consuming and increases the risk of overlooking critical clinical details.

To address these challenges, **Clinical Decision Support Systems (CDSS)** have been developed to assist clinicians in analyzing patient data and recommending **guideline-based treatment options**. CDSS platforms compare patient information against a structured knowledge base of clinical guidelines and decision rules, enabling more efficient and consistent treatment planning.

However, many existing oncology CDSS platforms rely heavily on **black-box machine learning models**, which lack **transparency and explainability**, thereby reducing clinician trust. Additionally, such systems often require **significant infrastructure and financial investment**, limiting their adoption in many Indian healthcare settings.

The **AI-Assisted Clinical Decision Support System** proposed in this project addresses these limitations by emphasizing **explainability, transparency, and accessibility within the Indian clinical context**. Instead of relying on opaque machine learning models, the system employs a **rule-based decision engine derived from established oncology guidelines**, ensuring that every recommendation is **traceable to specific clinical parameters and guideline logic**. Furthermore, the system integrates **Natural Language Processing (NLP)** techniques to extract structured clinical information from unstructured diagnostic reports, reducing manual effort and improving efficiency.

This project specifically focuses on the **Indian demographic**, where variations in **genetic factors, healthcare access, diagnostic practices, and treatment protocols** influence clinical decision-making. Consequently, the system is designed using **guideline adaptations and clinical practices relevant to India**, and its outputs are not intended to be directly generalized to other populations without appropriate validation. By integrating **automated data extraction, explainable reasoning, clinician oversight, and secure data management**, the proposed system aims to support oncologists in delivering **efficient, accurate, and patient-centered breast cancer care**.

PROBLEM STATEMENT

Breast cancer treatment planning is a complex clinical process that requires the integration of multiple diagnostic parameters derived from diverse sources of medical information. Oncologists must analyze **pathology reports, immunohistochemistry results, radiology findings, staging information, lymph node status, and patient health history** before determining an appropriate treatment strategy. The increasing complexity of modern oncology diagnostics has made this decision-making process highly **data-intensive and cognitively demanding**.

A key challenge in breast cancer management is the **fragmentation of patient data across multiple diagnostic documents**. Clinical information such as **tumor characteristics, biomarker status (ER, PR, HER2, Ki-67), lymph node involvement, and staging details** is often distributed across separate reports generated by different departments. Clinicians are required to manually consolidate this information to construct a comprehensive clinical profile, which can be both time-consuming and error-prone.

The manual interpretation of clinical reports introduces the risk of **overlooking critical indicators**, particularly in high-volume healthcare settings. For instance, misinterpretation or omission of **HER2 status, Ki-67 index, or lymph node involvement** can significantly impact treatment decisions. **Lymph node involvement**, in particular, is a crucial determinant of disease progression and treatment intensity, as it directly influences **cancer staging, recurrence risk, and the need for systemic therapies such as chemotherapy and radiation**.

Another significant challenge is the **rapid evolution of clinical oncology guidelines**. Organizations such as **NCCN, ESMO, and ASCO** frequently update their treatment recommendations based on emerging clinical evidence. Keeping up with these updates requires continuous effort from clinicians, particularly in resource-constrained and high-patient-load environments such as those commonly found in India. Additionally patients often face difficulty understanding their diagnosis and treatment plans. Breast cancer treatment typically involves multiple stages, including **surgery, chemotherapy, radiation therapy, targeted therapy, and endocrine therapy**. The use of complex medical terminology in clinical reports can lead to confusion, anxiety, and reduced patient engagement.

Furthermore, the **security and privacy of patient data** is a critical concern. Healthcare systems must ensure that sensitive medical information is protected against unauthorized access, data breaches, and misuse. Any system designed to support clinical decision-making must therefore incorporate **robust security mechanisms**, including **encryption, authentication, and role-based access control**. Despite the availability of CDSS tools, there remains a lack of **accessible, explainable, and context-aware systems** tailored to the **Indian healthcare environment**. Existing systems often lack **transparency, integration with clinical workflows, and adaptability to region-specific practices**, limiting their effectiveness in real-world deployment.

Therefore, the problem addressed in this project is the absence of a **transparent, explainable, and secure clinical decision support system** capable of integrating fragmented diagnostic data, incorporating **biomarker-driven and lymph node-based clinical reasoning**, and supporting oncologists in planning breast cancer treatment pathways. The proposed solution aims to address this gap by combining **automated data extraction, explainable rule-based reasoning, clinical validation mechanisms, and strong data privacy protections** within a unified platform designed for practical use in the Indian healthcare context.

PROPOSED SOLUTION

The proposed solution is an **AI-Assisted Clinical Decision Support System (CDSS)** designed to support oncologists in planning treatment pathways for **breast cancer patients**. The system focuses on converting **unstructured clinical reports** into structured medical data and applying **guideline-based reasoning** to generate treatment recommendations that are transparent, explainable, and clinically safe.

At the core of the system is an **automated data extraction pipeline** powered by **Natural Language Processing (NLP)**. Diagnostic reports such as pathology reports, other test results, and staging summaries are uploaded into the system. These documents often contain large volumes of unstructured text describing tumor characteristics, receptor status, and other diagnostic findings. The NLP pipeline processes these reports to identify and extract key clinical indicators such as **tumor size, lymph node involvement, ER status, PR status, HER2 expression, Ki-67 proliferation index, and disease stage**.

The extracted information is converted into a **structured patient profile**, which represents the patient's oncological characteristics in a standardized format. Structured data enables the system to perform computational reasoning and evaluate treatment rules efficiently.

To ensure clinical safety, the system incorporates a **Clinical Data Validation Module**. This module verifies whether essential fields required for treatment planning are present. For example, biomarker information such as **HER2 status or lymph node count** is critical for determining treatment pathways. If any required field is missing or ambiguous, the system automatically generates a **flag for clinician review**. This ensures that incomplete or uncertain information does not lead to incorrect treatment recommendations.

After the patient data has been validated, the system performs **molecular subtype classification**. Breast cancer is commonly classified into molecular subtypes based on receptor status and proliferation markers. These subtypes include **Luminal A, Luminal B, HER2-enriched, and Triple-Negative Breast Cancer (TNBC)**. Each subtype has distinct treatment strategies and prognosis. The system applies classification rules derived from established oncology guidelines to determine the molecular subtype.

Once subtype classification is complete, the patient profile is evaluated by the **Rule-Based Decision Engine**. This engine contains a structured knowledge base of treatment rules derived from internationally recognized oncology guidelines, including **NCCN, ESMO, ASCO, and ISMPO clinical practice guidelines**. Each rule represents a specific clinical condition and associated treatment pathway.

For example, a simplified rule may be represented as follows:

If ER positive and HER2 negative with low Ki-67, classify as Luminal A and recommend endocrine therapy.

Similarly, treatment pathways may depend on **lymph node involvement**. For instance, Luminal A tumors that are node-negative may be treated primarily with endocrine therapy, whereas **node-positive Luminal A tumors may require additional chemotherapy** depending on clinical risk factors.

In addition to identifying treatment pathways, the system performs **Contraindication and Safety Analysis**. This module evaluates potential treatment risks based on patient health conditions. Certain chemotherapy drugs may pose risks for patients with cardiac disease or kidney dysfunction. If a treatment recommendation conflicts with the patient's medical condition, the system generates a **safety warning**. These warnings alert clinicians to potential complications and encourage careful evaluation before finalizing treatment plans.

Once treatment pathways have been identified and safety checks completed, the system presents the results through the **Doctor Dashboard**. This interface displays a comprehensive patient summary, including biomarker status, disease stage, and recommended treatment options. Each recommendation is accompanied by an explanation showing which clinical rules were triggered and which guideline supports the recommendation.

The system follows a **Human-in-the-Loop Design Principle**, meaning that the final treatment decision remains under the authority of the clinician. Doctors can review, modify, or override the system's recommendations. Any modifications made by clinicians are recorded in the system's **audit log** for transparency and accountability.

Given the sensitivity of medical information, the system integrates **robust security and privacy mechanisms**. These include **data encryption, secure authentication, role-based access control (RBAC), and audit logging**. The design follows **privacy-by-design principles inspired by HIPAA data protection standards**, ensuring that patient information remains secure throughout the system lifecycle.

By combining **automated data extraction, guideline-based reasoning, clinical validation, and secure architecture**, the proposed solution aims to support oncologists in delivering accurate, efficient, and patient-centered breast cancer treatment planning.

METHODOLOGY/BLOCK DIAGRAM

The proposed system follows a structured methodology consisting of multiple interconnected modules that process clinical data and generate treatment recommendations.

The workflow begins with the **Data Ingestion Phase**, where clinicians upload diagnostic reports into the system. These reports may include **pathology reports, test results, radiology summaries, or laboratory reports**. The system accepts multiple document formats, including **PDF, text files, and structured electronic health records**.

Once uploaded, the reports undergo **Preprocessing**. In this stage, the system converts document contents into machine-readable text using document parsing and optical character recognition if necessary. The text is then prepared for analysis through tokenization, normalization, and sentence segmentation.

Following preprocessing, the system executes the **Natural Language Processing (NLP) Pipeline**. This module uses a combination of **Named Entity Recognition (NER)** models and rule-based pattern matching to identify key clinical entities. Important terms such as **ER status, HER2 status, tumor size, lymph node involvement, and disease stage** are extracted from the report text.

After data extraction, the system performs **Clinical Data Validation**. This step verifies whether all required clinical indicators are available. If essential fields are missing, the system generates a **clinical validation alert** and requests manual confirmation from the clinician.

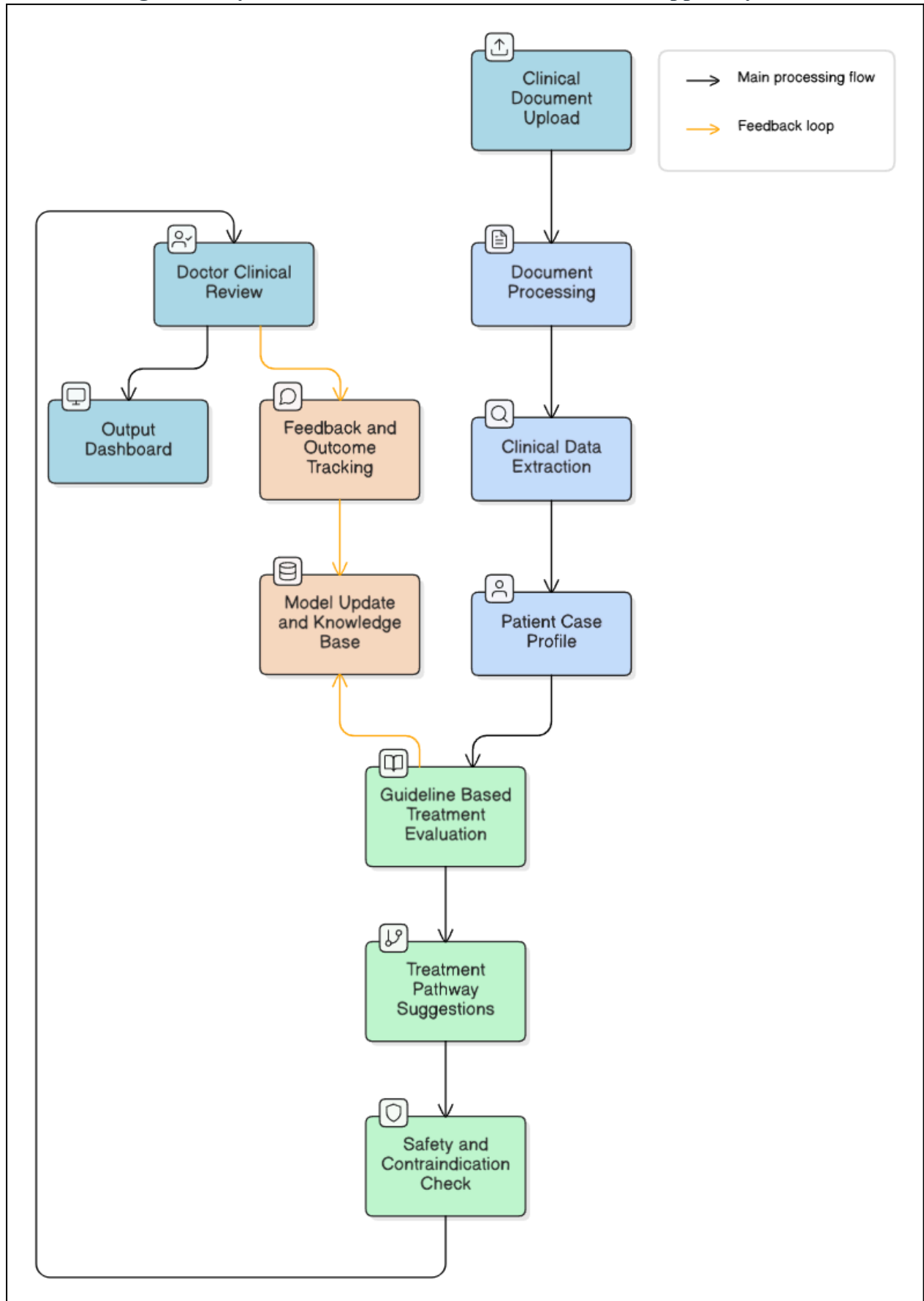
Validated data is then processed by the **Subtype Classification Module**. Using biomarker-based rules, the module determines the molecular subtype of the tumor. The subtype classification result forms the basis for selecting appropriate treatment pathways.

Next, the structured patient profile is evaluated by the **Rule-Based Decision Engine**. The engine compares the patient data against a set of **guideline-derived decision rules**. These rules represent treatment recommendations established by oncology organizations. Each rule consists of a clinical condition and an associated treatment recommendation.

The output of the decision engine is passed to the **Contraindication and Safety Module**, which analyzes potential treatment risks. If a treatment recommendation conflicts with patient comorbidities or laboratory values, the system generates a **safety alert** for clinician review.

Finally, the treatment recommendations are presented through the **Doctor Dashboard Interface**. Clinicians review the recommendations and either approve or modify the suggested treatment plan.

Figure 1 – System Workflow of the Clinical Decision Support System



Architecture Overview for figure 1

The proposed system is an AI-based clinical decision support system designed to assist in personalized cancer treatment planning. The process begins with the upload of clinical documents such as pathology reports, biomarker reports, and patient health records. These documents are processed to extract relevant textual information through parsing and data cleaning.

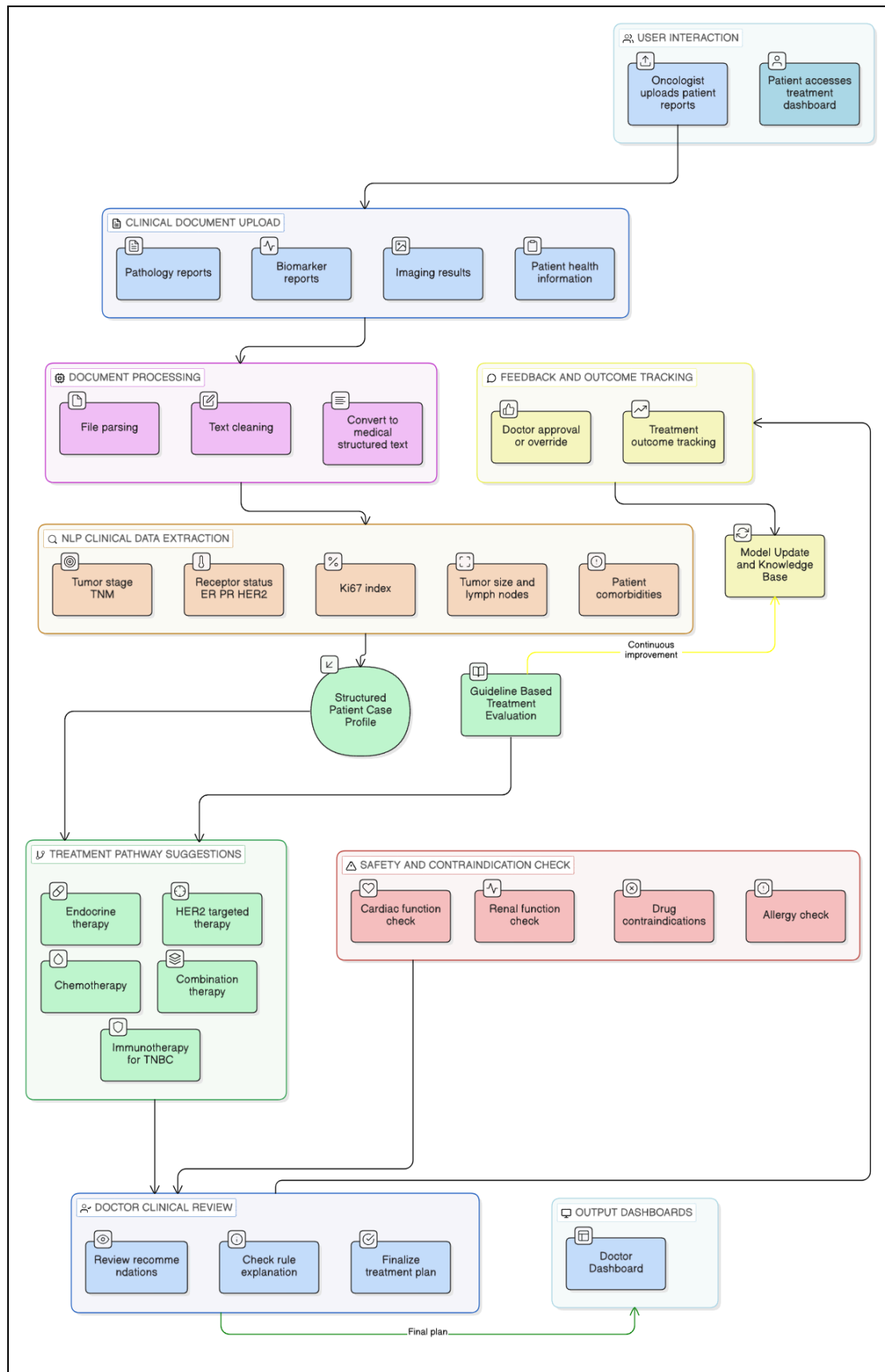
Using Natural Language Processing (NLP) techniques, important clinical features such as tumor stage, biomarker status (HER2, ER, PR), tumor size, and patient comorbidities are identified and converted into a structured patient case profile. This structured data serves as the input for the decision-making module.

The system then evaluates the patient profile using a guideline-based treatment evaluation engine aligned with standard clinical protocols. Based on this evaluation, it generates appropriate treatment pathway suggestions, such as endocrine therapy, HER2-targeted therapy, or chemotherapy. These suggestions are further validated through safety and contraindication checks to ensure patient-specific suitability.

The recommendations are then reviewed by the doctor, who can approve, modify, or override them. The final treatment plan is displayed through the output dashboard.

Additionally, the system incorporates a feedback and outcome tracking mechanism, where doctor decisions and treatment outcomes are recorded. This feedback is used to update the model and knowledge base, enabling continuous improvement and more accurate recommendations over time.

Figure 2 – System Workflow of the Clinical Decision Support System



Architecture Overview for figure 2

The system architecture provides a comprehensive workflow for processing clinical data and generating personalized cancer treatment recommendations. It begins with the user interaction layer, where the oncologist uploads patient reports and the patient can access the treatment dashboard.

The uploaded clinical documents, including pathology reports, biomarker reports, imaging results, and patient health information, are collected in the clinical document upload module. These documents are then processed through a document processing pipeline, which involves file parsing, text cleaning, and conversion into structured medical text.

Next, the system performs NLP-based clinical data extraction to identify key medical features such as tumor stage (TNM), receptor status (ER, PR, HER2), Ki-67 index, tumor size, lymph node involvement, and patient comorbidities. These extracted features are combined to form a structured patient case profile.

The patient profile is then evaluated using a guideline-based treatment evaluation module, which follows established clinical protocols to determine suitable treatment strategies. Based on this evaluation, the system generates treatment pathway suggestions such as endocrine therapy, HER2-targeted therapy, chemotherapy, combination therapy, and immunotherapy for specific cases like TNBC.

Before presenting the recommendations, a safety and contraindication check is performed to ensure that the suggested treatments are compatible with the patient's condition. This includes checks for cardiac and renal function, drug contraindications, and allergies.

The final recommendations are reviewed by the doctor in the clinical review module, where the doctor can validate, modify, and finalize the treatment plan. The approved plan is then displayed through the output dashboard.

Additionally, the system includes a feedback and outcome tracking mechanism, where doctor approvals or overrides and treatment outcomes are recorded. This data is used to update the model and knowledge base, creating a continuous improvement loop that enhances the system's performance over time.

Figure 3 – Biomarker-Driven Breast Cancer Classification Algorithm

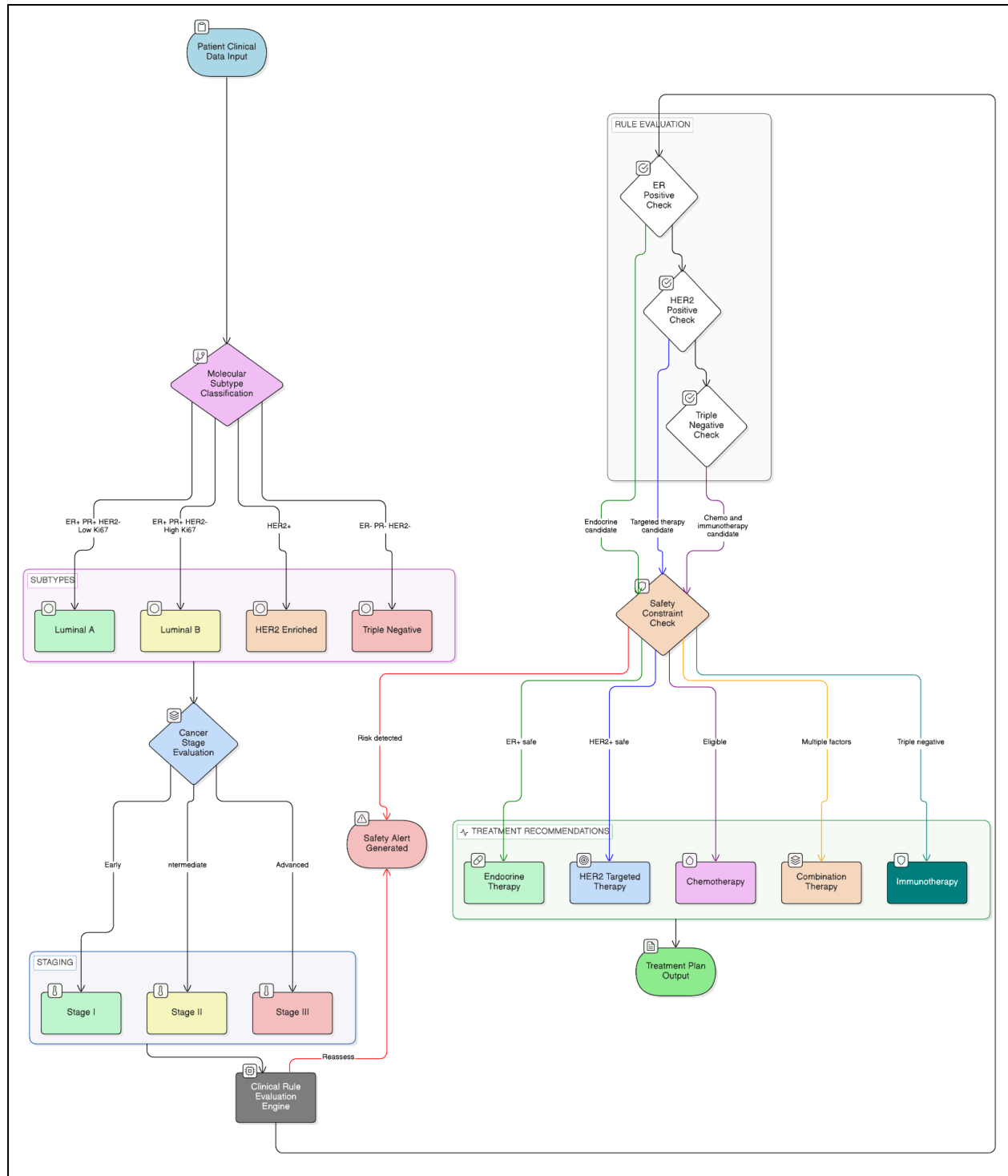
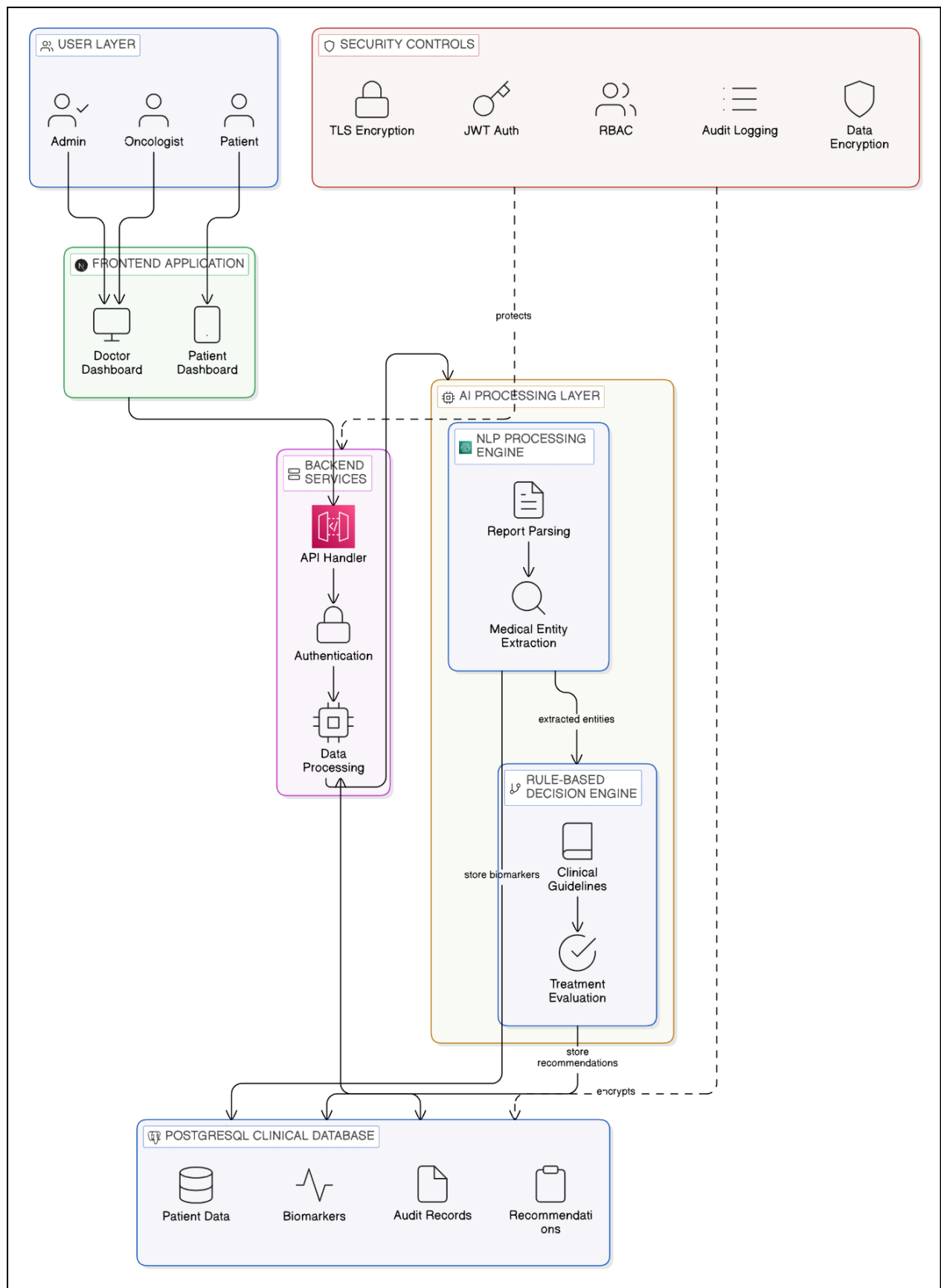


Figure 4 – System Architecture with Security and Data Protection Layers



HARDWARE AND SOFTWARE REQUIRED

The development and deployment of the proposed system require both hardware infrastructure and specialized software components.

Hardware Requirements

The system can be developed and tested on a standard personal computer or workstation. Recommended specifications include a multi-core processor, a minimum of **8 GB RAM**, and sufficient storage capacity for document processing and database operations. Internet connectivity is required for cloud deployment and collaboration.

Software Requirements

The frontend interface is implemented using **Next.js**, a modern web development framework based on **React**. Next.js supports server-side rendering and provides efficient performance for interactive web applications.

The backend server is implemented using **FastAPI**, a high-performance Python framework designed for building scalable web APIs. FastAPI provides automatic API documentation and supports asynchronous processing, which is beneficial for handling multiple concurrent requests.

The system uses **PostgreSQL** as the primary database management system. PostgreSQL provides strong reliability, advanced query capabilities, and support for structured data storage.

The Natural Language Processing module is implemented using **Python libraries such as spaCy**, combined with rule-based extraction methods for clinical terminology. This hybrid approach helps reduce NLP errors while maintaining high extraction accuracy.

Development tools include **Visual Studio Code for programming, GitHub for version control, and Docker for containerized deployment**. Containerization ensures consistent system behavior across different environments.

Security Tools

Security features are implemented using **TLS encryption for secure data transmission, AES-256 encryption for stored data, and JWT-based authentication mechanisms**. Role-based access control ensures that users can only access information relevant to their role.

PROPOSED EVALUATION MEASURES

The effectiveness of the system will be evaluated through multiple evaluation criteria covering **technical performance, clinical accuracy, usability, and security**.

NLP Extraction Accuracy

The Natural Language Processing module will be evaluated using standard metrics including **precision, recall, and F1-score**. These metrics measure the system's ability to correctly extract clinical entities from diagnostic reports.

Clinical Decision Accuracy

The rule-based decision engine will be validated using a set of simulated breast cancer case scenarios. System recommendations will be compared against treatment guidelines provided by oncology organizations. The level of agreement between system outputs and guideline recommendations will be measured.

System Usability

Usability will be evaluated using the **System Usability Scale (SUS)** questionnaire. Clinicians will interact with the doctor dashboard and provide feedback on ease of use, interface design, and workflow integration.

Security and Privacy Testing

Security evaluation will verify the implementation of encryption protocols, authentication mechanisms, and role-based access control. The system will be tested for vulnerabilities such as unauthorized data access or insecure communication channels.

System Performance and Clinical Validation

System performance will be evaluated based on **response time, throughput, and reliability** under simulated clinical workloads to ensure the platform remains responsive during real-world usage.

In addition to technical performance, **clinical decision accuracy** will be assessed through **validation by practicing oncologists**. System-generated treatment recommendations will be compared with **doctor opinions and guideline-based decisions** to measure correctness and clinical relevance.

The system will also incorporate a **doctor feedback mechanism**, where clinicians can review, accept, modify, or reject recommendations. This feedback will be recorded and used to evaluate system effectiveness and identify areas for improvement.

Over time, the system can evolve using a **learning-based improvement approach**, where feedback from previous cases and outcomes of subsequent patient analyses are used to refine decision rules. This creates a **reinforcement learning-inspired feedback loop**, enabling continuous enhancement of system performance while maintaining clinician oversight.

CONCLUSION

The proposed system demonstrates a structured and scalable approach to handling complex clinical data derived from oncology guidelines such as NCCN and ESMO. Unstructured medical documents are effectively processed using Natural Language Processing techniques, enabling the extraction of key clinical entities including biomarkers (ER, HER2, Ki-67), nodal status, comorbidities, and treatment indicators. Data preprocessing steps such as text cleaning, normalization, and entity mapping ensure that noisy and fragmented clinical text is transformed into a consistent and machine-readable format.

The algorithmic framework combines rule-based reasoning with contextual understanding through ClinicalBERT embeddings, allowing the system to capture both explicit guideline logic and implicit semantic relationships within clinical text. A priority-based decision engine ensures clinically relevant and conflict-free recommendations by considering multiple patient attributes simultaneously. Furthermore, the integration of comorbidity and contraindication handling enhances the safety and real-world applicability of the system.

An important strength of the proposed approach lies in its adaptability. The feedback-driven learning mechanism enables continuous refinement of rules based on clinician input, thereby improving recommendation accuracy over time. This hybrid architecture combining deterministic rules, machine learning representations, and human-in-the-loop feedback provides a robust foundation for building explainable and clinically assistive decision support systems.

Overall, the system effectively bridges the gap between unstructured medical knowledge and actionable clinical insights, offering a reliable, interpretable, and extensible solution for AI-assisted healthcare decision-making.

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NCCN Breast Cancer Guidelines

<https://www.nccn.org/guidelines/guidelines-detail?category=1&id=1419>

ESMO Breast Cancer Clinical Practice Guidelines

<https://www.esmo.org/guidelines/breast-cancer>

ASCO Breast Cancer Treatment Guidelines

<https://ascopubs.org/topics/asco-guidelines/breast-cancer>

TCGA Breast Cancer (BRCA) Dataset

<https://portal.gdc.cancer.gov/projects/TCGA-BRCA>

The Cancer Imaging Archive (TCIA) Breast Cancer Imaging Collections

<https://www.cancerimagingarchive.net/browse-collections/>

VinDr-Mammo Mammography Dataset

<https://physionet.org/content/vindr-mammo/1.0.0/>

Breast Cancer Histopathology Slide Dataset

<https://zenodo.org/record/7081149>